AccuKnox

Secure Code to Cognition™

Onprem Airgapped Container Security PoC

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Outline

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1. Deployment Architecture
2. Pre-Requisites
3. POC Scenarios, Success Criteria
4. POC Status/Timelines
5. Integrations (Notification Channels, Ticketing)
6. Executive Report & Summary

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Container Security

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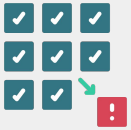
Security Monitoring

- Threat Detection & Response
- Anomaly Detection

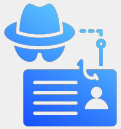
Security Enforcement

- Preemptive Controls
- File | Process | Network



Hard Isolation

- Multi Tenant Deployment risks
- Hard Isolation



Identity & Access Risks

- Overprivileged Access
- Unused Accounts

Vulnerability Detection

- Container Image Vulnerabilities
- Sensitive Data Scan

Admission Controller

- Deployment from allowed registries
- Adm Security Best Practices

Deployment Types

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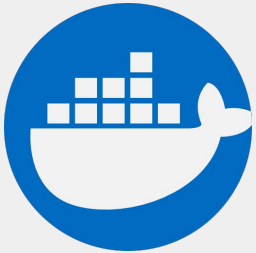
Kubernetes

- Robin Clusters
- K8s onprem & managed



Containers

- Containers on VM, ECS
- Docker, Containerd, CRIO, Podman

VM / Bare-Metal

- Virtual Machines
- Bare-Metal

IoT / Edge

- ARM, x86
- Lightweight Edge

POC Deployment Architecture

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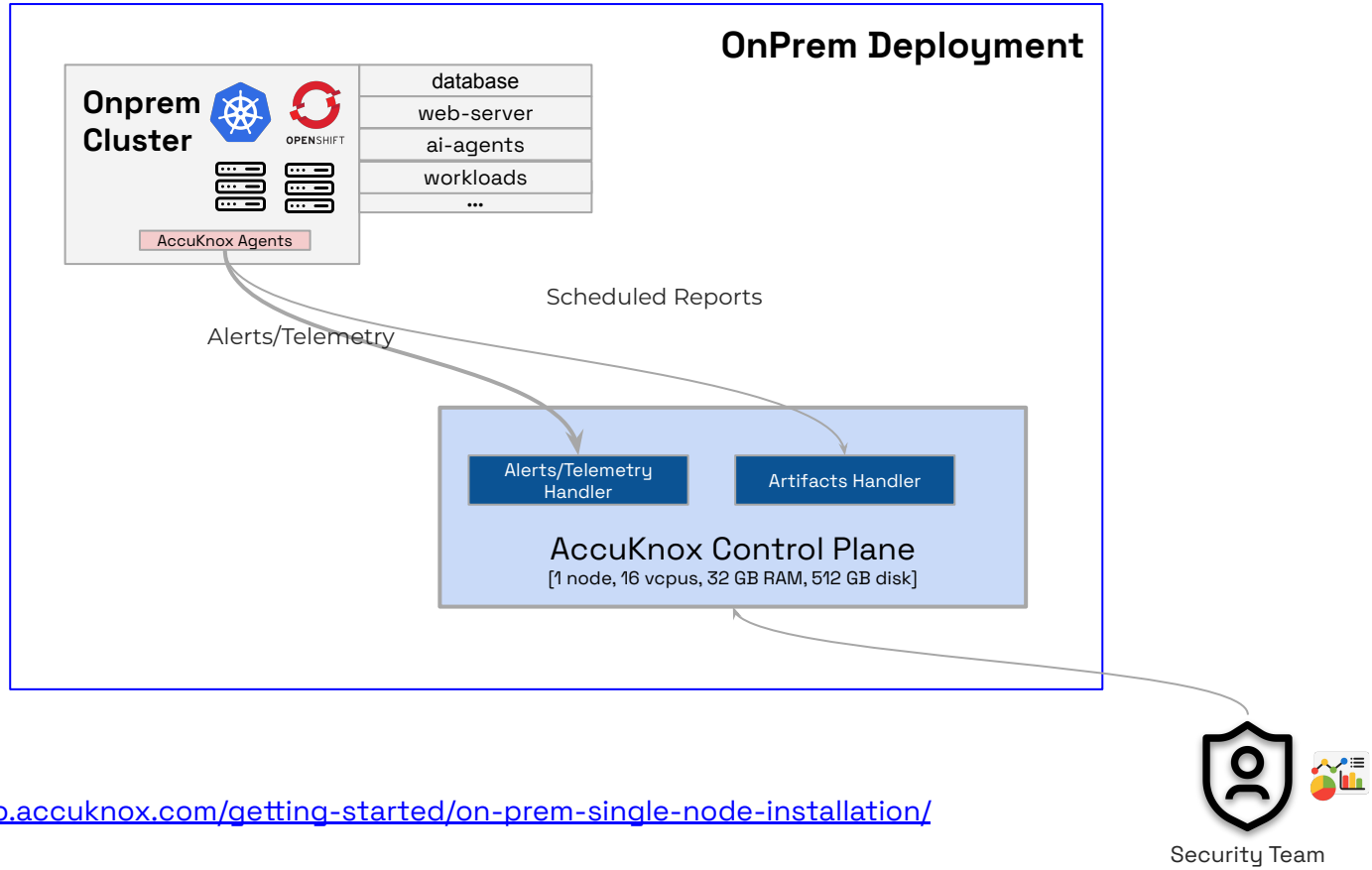

Ref: https://help.accuknox.com/getting-started/on-prem-single-node-installation/

POC Use-Cases (Container Runtime Security)

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Use-Case	Success Criteria	Status
File Integrity Monitoring/Protection	1. Updating system's bin folders or boot folders should be "Permission Denied". 2. Violation should be reported in AccuKnox Control Plane.	
Protecting Trusted Root Certificates and certificate bundle tampering	1. Any update, deletion, or modification of trusted root certificate stores and certificate bundle directories (e.g., *cert*, *ca*, *bundle*) should be blocked. 2. Violation should be reported in AccuKnox Control Plane.	
Flag Process executions from /tmp folder	1. Execution of binaries or scripts from /tmp should be blocked or flagged based on policy. 2. Violation should be reported in AccuKnox Control Plane with all necessary details.	
Detect Anomalous Process Executions from Application	1. Policies should be auto-discovered based on application runtime behavior. 2. Execution outside the discovered baseline should be flagged as a violation when enforced. 3. Violation should be reported in AccuKnox Control Plane with the ability to update the policy or discard the change.	
Prevent from Crypto Mining executions	1. Execution of known crypto-mining binaries or processes should be blocked at runtime. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Prevent unauthorized access to environment	1. Access to environment variables should be blocked by default. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Audit use of package management tools at runtime	1. Runtime use of package management tools (e.g., apt, dnf) should be audited. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Prevent Container Management tools from getting used from within containers	1. Runtime execution of container management tools (e.g., docker, crictl, kubectl) from within containers should be blocked. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Prevent use of mascan, nmap from with containers	1. Runtime execution of network scanning tools (e.g., masscan, nmap) from within containers should be blocked. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	

Container Runtime Security (contd)

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Use-Case	Success Criteria	Status
Prevent/Audit file system related operations (mount, nsenter, fsck, lsblk, ...)	1. Runtime execution of file system-related operations (e.g., mount, nsenter, fsck, lsblk) should be prevented or audited based on policy. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Prevent use of Reconnaissance or Penetration Testing tools from inside containers	1. Runtime execution of reconnaissance or penetration testing tools from within containers should be blocked. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Ability to apply policies from cli or from control plane or using gitops	1. Security policies should be applicable via CLI, AccuKnox Control Plane, or GitOps workflows. 2. Policy changes should be versioned and reflected consistently across the environment.	
Classify process executions from terminal or background processes	1. Runtime process executions should be classified based on origin (interactive terminal vs background/system). 2. Classification results should be visible in AccuKnox Control Plane.	
Ability to see the policy violations from the control plane for audit/block events	1. All audit and block events should be centrally visible in AccuKnox Control Plane. 2. Events should clearly indicate policy action taken (audit or block).	
Protect Hashicorp Vault Secrets Manager onprem deployment (Advanced)	1. Unauthorized access or interaction with the on-prem secrets manager should be blocked at runtime. 2. Violations should be reported in AccuKnox Control Plane with all necessary details.	
Handle trigger points when an audit/block event is detected and send it to notification channel	1. Audit or block events should trigger configurable actions. 2. Events should be forwarded to configured notification channels.	
Process Whitelisting: Zero Trust policy allowing execution of certain processes only and denying rest.	1. Under a Zero Trust execution model, only explicitly allow-listed processes should be permitted to execute at runtime. 2. Execution of all other processes should be blocked and reported in AccuKnox Control Plane.	
Nano Segmentation: Only allow trusted processes to handle network communication. Dynamically	1. Only trusted processes should be allowed to initiate or handle network communication at runtime. 2. Trusted processes should be dynamically identified based on observed application behavior. 3. Unauthorized network access attempts should be blocked and reported in AccuKnox Control	

Use cases

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Category	Use-Case	Success Criteria	Status
Compliance	CIS scan of kubernetes cluster	1. Kubernetes clusters should be scanned against CIS benchmarks. 2. Findings should be reported in AccuKnox Platform with compliance status and failed controls.	
	MITRE Attack framework mapping for the policies and policy violations	1. Security policies and detected violations should be mapped to relevant MITRE ATT&CK techniques. 2. Mappings should be visible in AccuKnox Platform.	
	NIST800-53 mapping	1. Security controls and findings should be mapped to NIST 800-53 requirements. 2. Mappings should be available in AccuKnox Control Plane for audit readiness.	
Cluster Misconfiguration Detection	Identify service accounts with anonymous access enabled	1. Service accounts with anonymous or unauthenticated access should be identified. 2. Findings should be reported in AccuKnox Control Plane.	
	Identify Secrets / credentials in manifests and configuration files	1. Kubernetes manifests and configuration files should be scanned for embedded secrets or credentials. 2. Violations should be reported in AccuKnox Platform.	
	Ensure CPU/Memory limits are set across all workloads	1. Workloads missing CPU or memory limits should be identified. 2. Findings should be visible in AccuKnox Control Plane.	
	Identify external facing workloads	1. Workloads exposed externally should be identified and listed. 2. Findings should be reported in AccuKnox Control Plane	
	Detect workloads/accounts with Host PID/IPC privileges	1. Workloads or service accounts using host PID or IPC namespaces should be detected. 2. Findings should be reported in AccuKnox Control Plane.	
	Detect use of privileged containers	1. Workloads running with privileged container settings should be detected. 2. Findings should be reported in AccuKnox Control Plane.	
	Detect use of writable hostpath volumes	1. Workloads using writable hostPath volumes should be detected. 2. Findings should be reported in AccuKnox Control Plane.	

Use cases

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Category	Use-Case	Success Criteria	Status
K8s Identity & Entitlements (KIEM)	Identify users, services, namespaces having ClusterAdmin privileges	1. Users, services, and namespaces with ClusterAdmin or equivalent privileges should be identified and listed. 2. Findings should be visible in AccuKnox Control Plane.	
	Identify k8s service accounts with elevated permissions	1. Kubernetes service accounts with elevated or excessive permissions should be identified. 2. Findings should be reported in AccuKnox Control Plane.	
	Identify unused roles not mapped to any workloads currently	1. Kubernetes roles not currently mapped to active workloads should be identified. 2. Findings should be visible in AccuKnox Control Plane.	
	Identify Subjects with Permission to create roles and rolebindings	1. Subjects with permissions to create or modify roles and role bindings should be identified. 2. Findings should be reported in AccuKnox Control Plane.	
Vulnerability Scanning	Scan the container images in the k8s environment	1. Container images used in the Kubernetes environment should be scanned for vulnerabilities. 2. Findings should be reported in AccuKnox Control Plane.	
	If a new critical vulnerability is detected, send an alert to the team (email, notification, webhook)	1. When a new critical vulnerability is detected, an alert should be generated. 2. Alerts should be sent to configured channels (email, notifications, webhooks).	
	Show vulnerabilities in the context of k8s clusters, namespaces, workloads	1. Vulnerabilities should be visible in the context of clusters, namespaces, and workloads. 2. Contextual views should be available in AccuKnox Platform.	
	Triage and handle findings lifecycle	1. Users should be able to triage vulnerability findings (review, suppress, accept risk, remediate). 2. Finding status and lifecycle updates should be tracked in AccuKnox Platform.	

POC Plan

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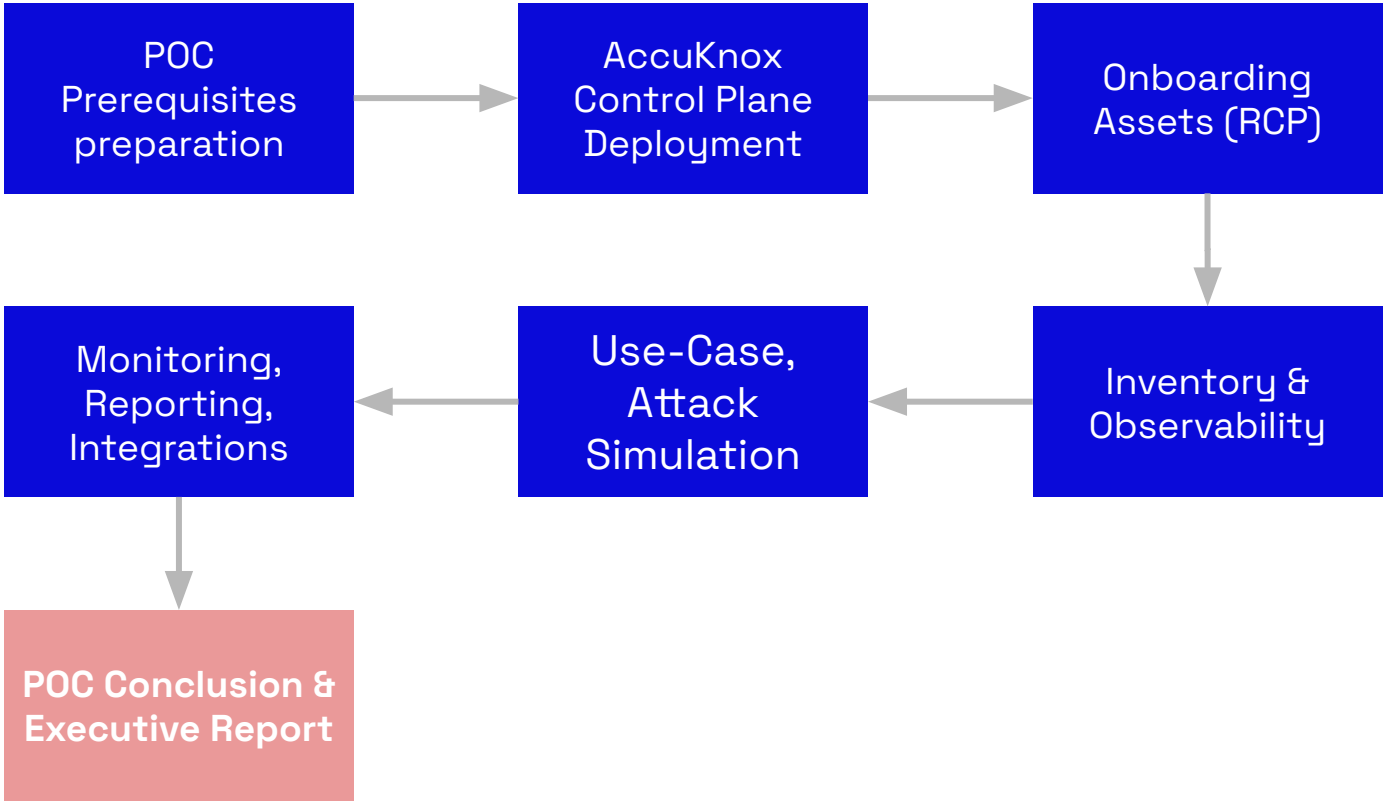

Week 1

Week 2

POC Prerequisites

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AccuKnox Control Plane Hardware Prerequisite

- 1 VMs, each with >=16 vCPUs, >=64GB RAM, >= 512 GB Disk
- 256 GB disk should be allocated to /var partition
- AccuKnox team will provide a licensed installation tgz (~20GB)
- No external network connectivity expected during installation

References

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- AccuKnox Enterprise Guide
- CWPP Security Use-cases
- CWPP Playbook
- KSPM Playbook

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Thank You

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